

PROJECT EXO 2020

SMART IV DRIP MONITORING USING IOT

TEAM NAME: SMARTIV DYNAMICS

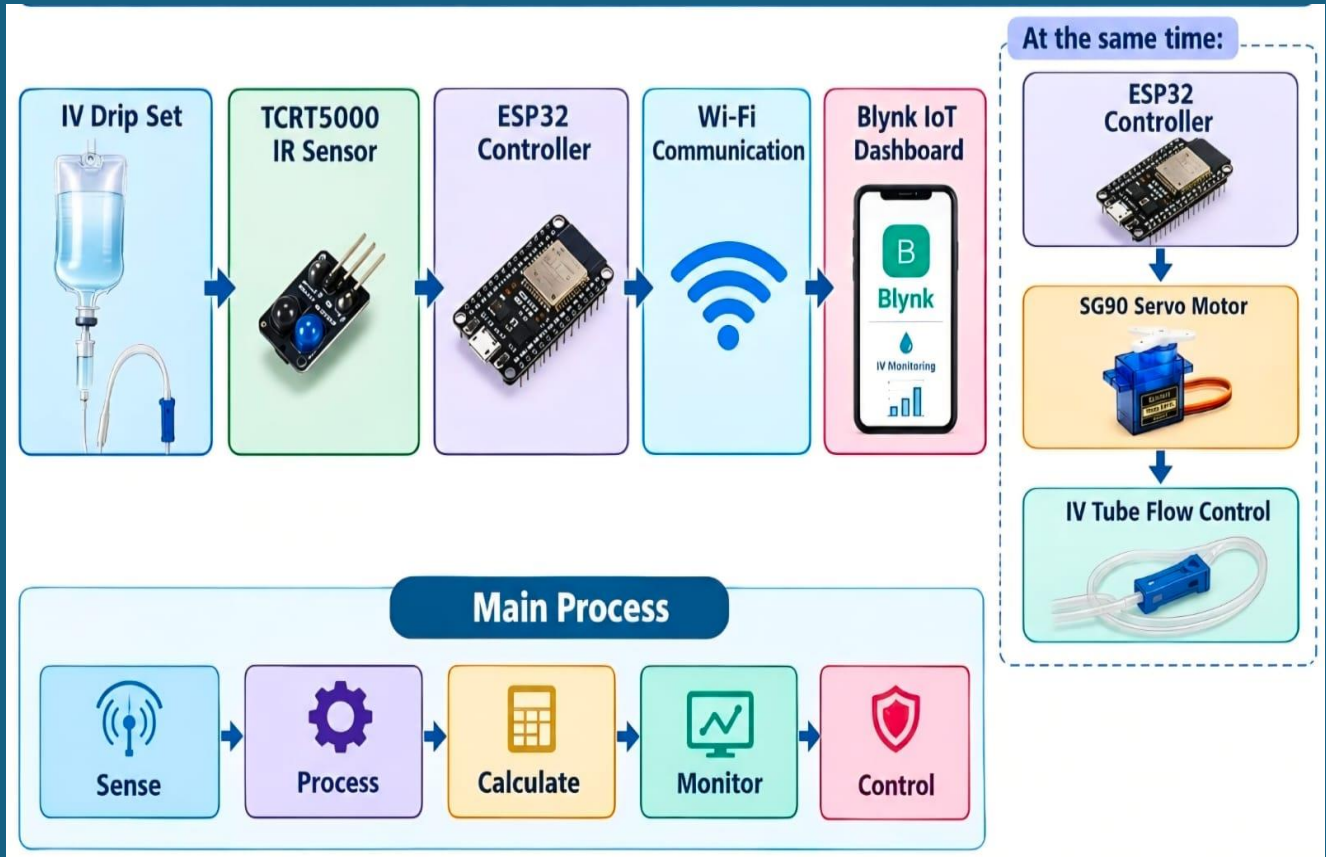
PROBLEM STATEMENT

- Manual IV drip monitoring requires frequent observation.
- Changes in drip flow may not be noticed immediately.
- Delayed detection can affect the intended delivery rate.
- An IoT-based prototype can provide real-time monitoring and remote visibility.

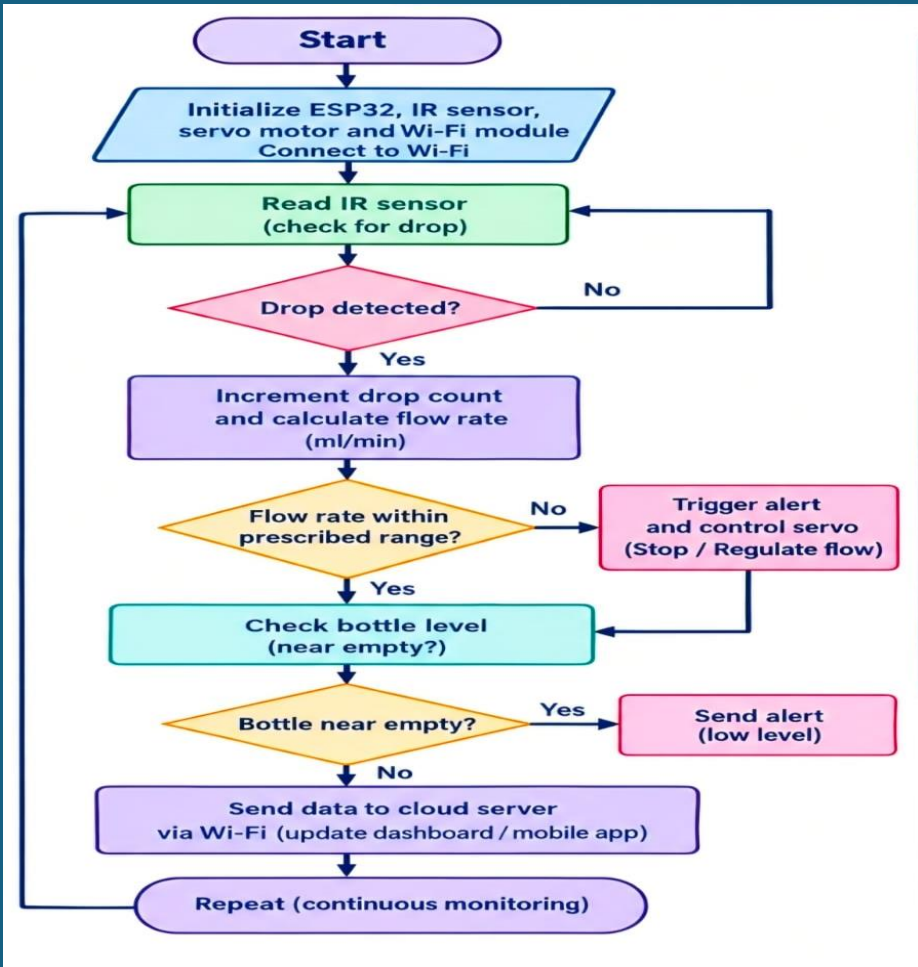
OBJECTIVES

- Detect individual IV drops in real time.
- Estimate flow rate in mL/min.
- Send monitoring data to the Blynk IoT dashboard.
- Alert when flow conditions deviate from the expected range.
- Demonstrate servo-based IV tube regulation.

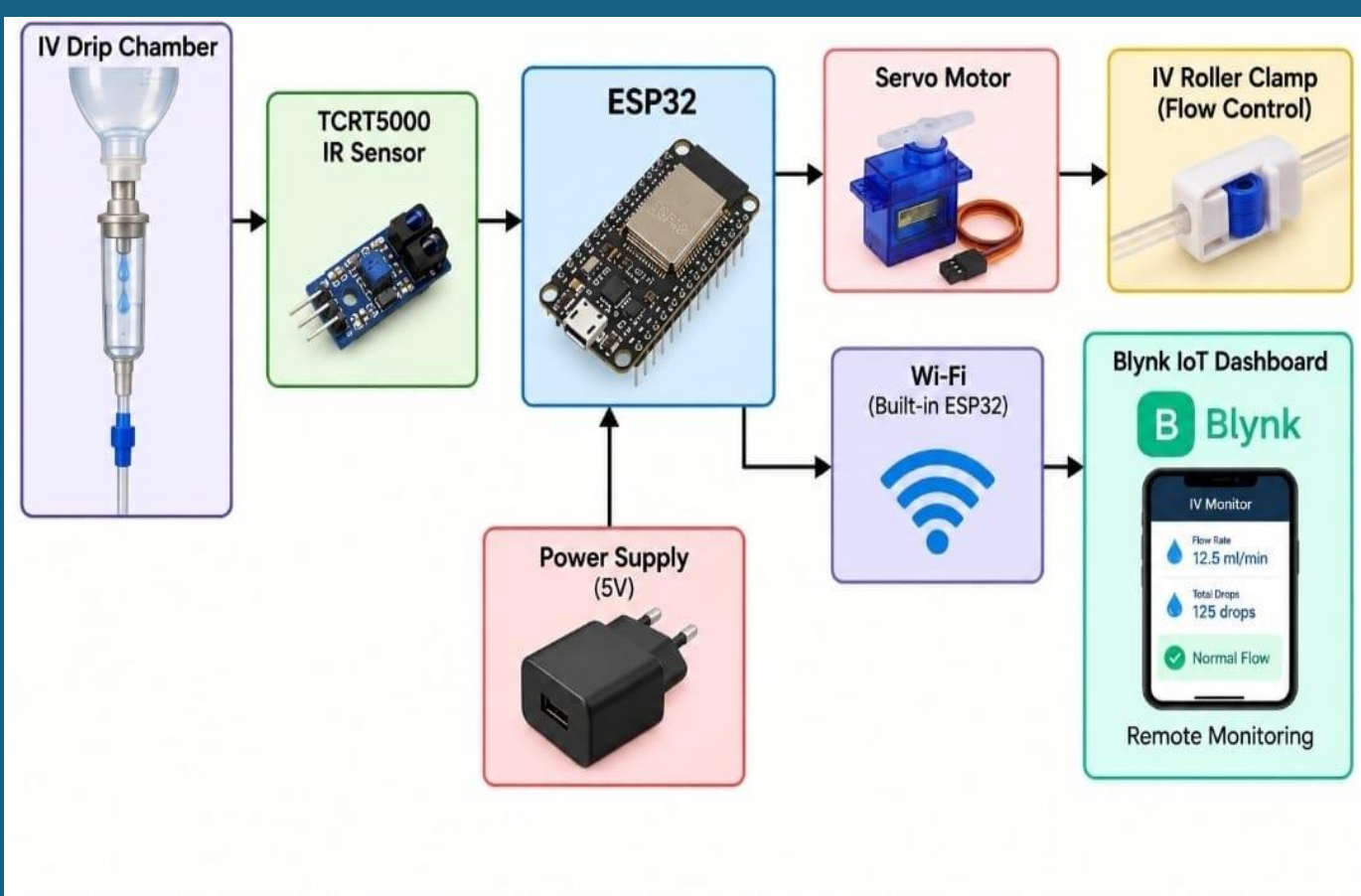
SYSTEM ARCHITECTURE



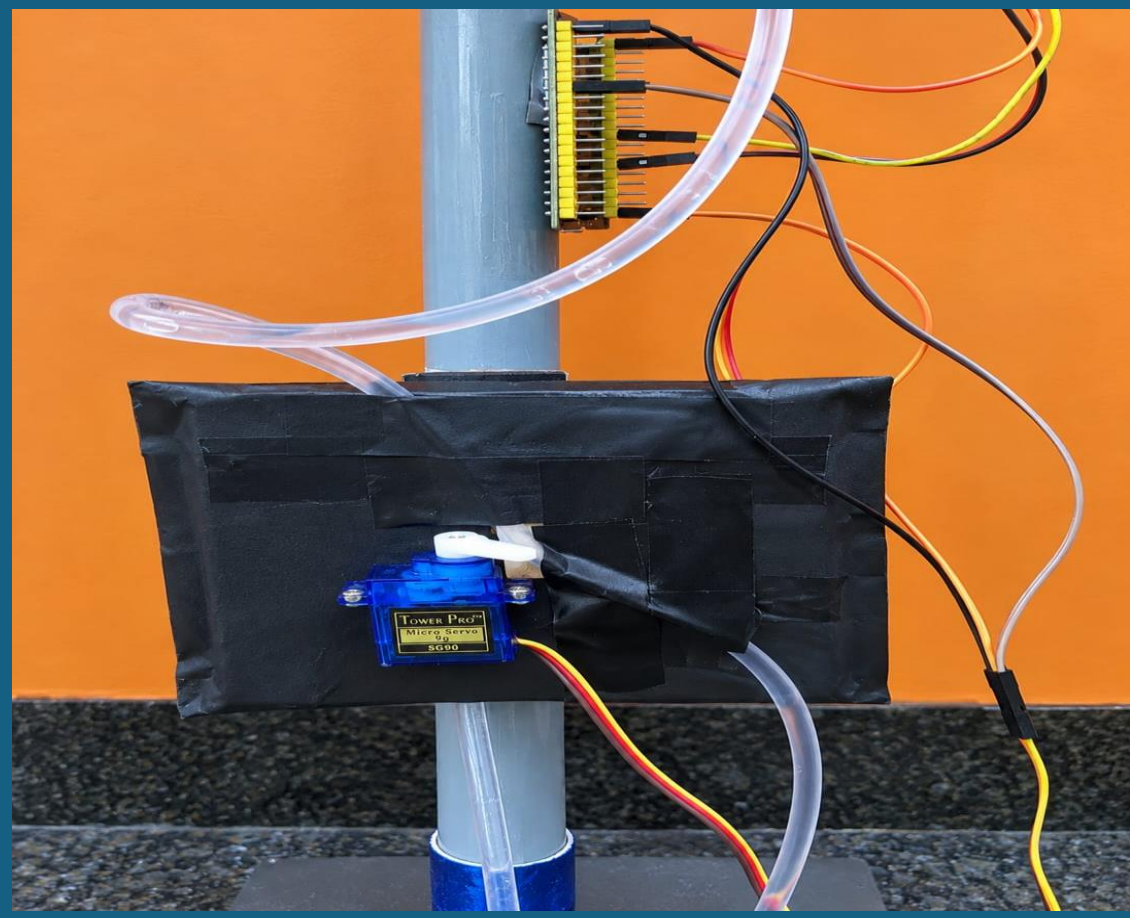
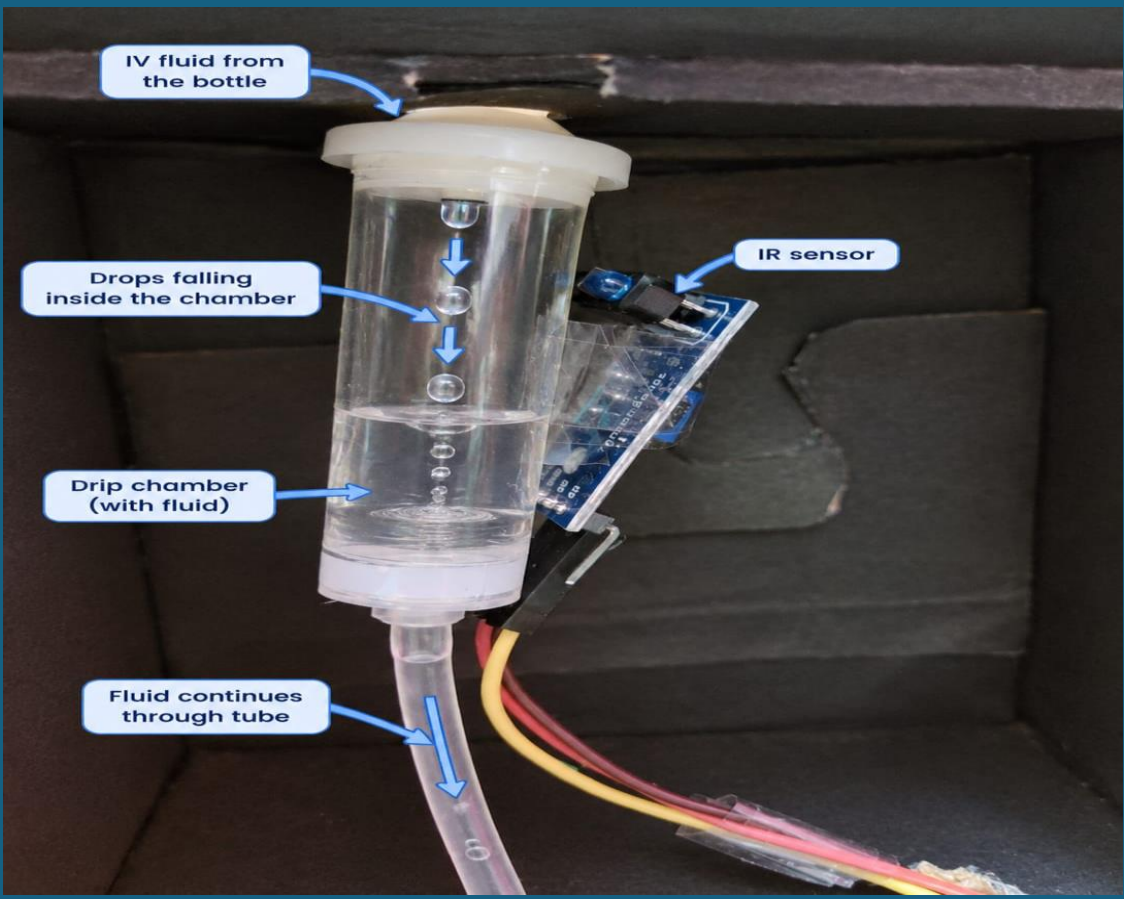
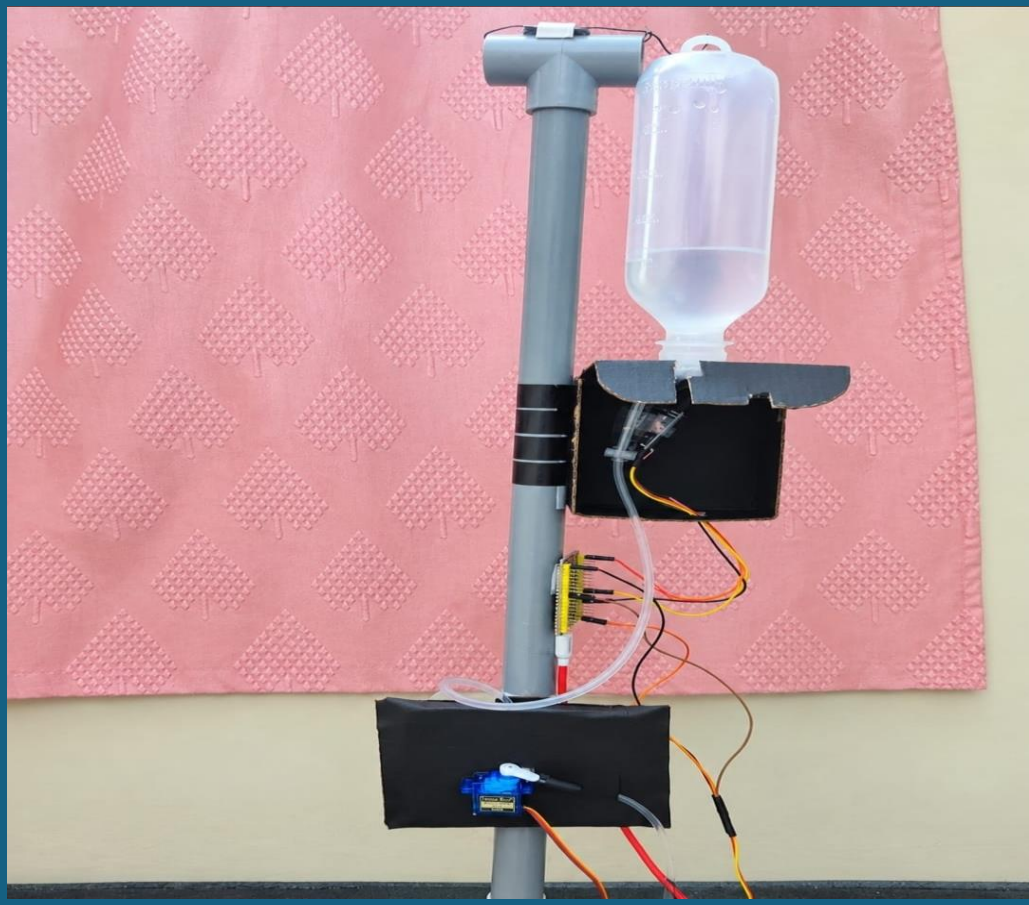
FLOW CHART



CIRCUIT DIAGRAM



PROTOTYPE



RESULT AND DISCUSSION

- ✓ Individual IV drops detected using TCRT5000
- ✓ Real-time signal processing using ESP32
- ✓ Approximate flow rate calculated from drop detection
- ✓ Data displayed remotely through Blynk IoT
- ✓ SG90 servo demonstrated for tube regulation
- ✓ Prototype provides a basis for improved automated IV monitoring

KEY FEATURES

- Real-time drip monitoring
- IoT-enabled remote dashboard
- Individual drop detection
- Flow-rate estimation
- ESP32-based system
- Servo-assisted tube regulation
- Remote monitoring

APPLICATIONS

- Hospital IV monitoring prototypes
- Patient-ward monitoring
- Remote healthcare monitoring
- Nursing assistance
- Smart healthcare research

CONCLUSION

The prototype demonstrates an IoT-based approach to monitor IV drip flow using ESP32 and TCRT5000. The system detects individual drops, estimates flow rate and displays monitoring information through the Blynk platform. A servo-based mechanism is also incorporated for prototype tube regulation.

FUTURE SCOPE

- Improve flow-rate measurement accuracy.
- Add reliable bottle-level detection.
- Improve the servo-based tube-control mechanism.
- Develop a dedicated mobile application.
- Integrate cloud/database monitoring.
- Perform controlled validation before practical medical use.

REFERENCES

- ESP32 Documentation
- Blynk IoT Documentation
- TCRT5000 Sensor Datasheet
- SG90 Servo Motor Datasheet
- Relevant research papers on IoT-based IV monitoring

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scan to view project repository

